the average age and the corresponding average income of the customer, one can use the following query:

```
select round(avg(Age),2)  as Age_bar,
round(avg(Annualincome),2) as Income_bar
from customer
```

Age_bar	Income_bar
38.95	60.79

Similarly, it is also possible to get the maximum and minimum ages as well as their corresponding income values. Median and mode features are not available in SQL and can be derived from other queries. In the earlier article, I have described the methods in detail.

Frequency distribution: Preparing the frequency table is an essential part of descriptive statistics and is also used for histogram plotting. It can be implemented simply by the *count()* function and the GROUP BY clause. To demonstrate its use, I have taken the 'order detail' table of the Northwind database to show the frequency of different products in the TABLE. The corresponding SQL statement is as follows:

```
USE northwind;
SELECT ProductID,COUNT(ProductID) AS Product-Frequency
     FROM `order details` AS a
GROUP BY ProductID;
```

ProductID	Product_ frequency
1	38
2	44
3	12
4	20
…	….

A little improvement in the above query can provide the percentage of the product count along with the frequency values. To do so, we require a sub-query to count the number of records for each category of ProductID. The query may be written as follows:

```
SELECT ProductID,
COUNT(ProductID) AS Product_Frequency,
ROUND(COUNT(ProductID)*100/(select COUNT(*) from `order
details`),2) AS Percentage
FROM `order details` AS a
GROUP BY ProductID;
```

ProductID	Product_Frequency	Percentage
1	38	1.76
2	44	2.04
3	12	0.56
4	20	0.93
5	10	0.46
…	…	…

Linear regression

Linear regression assumes that there exists a linear relationship between the response variable and the explanatory variables, and one can fit a line between the two (or more variables). The inclined line makes an angle m with the x-axis and intercepts y at c. The line is expressed as:

```
y = mx + c                         (1)
```

The slope angle with respect to the x-axis is expressed as m and the y-intercept is expressed as c.

If the sample size is n and there are two linearly dependent variables x and y, the slope angle is expressed in terms of x and y as:

$$x_i - \acute{x}$$
$$\frac{\sum (¿)(y_i - \acute{y})}{\sum (x_i - \acute{x})^2}$$
$$m = ¿$$

Figure 1: Plotting a car's age versus its price